

Carlo M. Vilches

Économiste — Marchés de l'énergie, régulation & modélisation quantitative

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Nationalité espagnole · mobile

PROFIL

Économiste (10+ ans) spécialisé dans la régulation sectorielle, la concurrence et la conception tarifaire, habitué à défendre des intérêts stratégiques devant les autorités de régulation. Je construis des modèles économiques de prévision et d'évaluation d'impact, je chiffre la compensation des obligations de service public et des dispositifs de subvention, et je communique les résultats de façon claire à la direction et aux décideurs institutionnels. Maîtrise des outils statistiques et économétriques (Python, R) et d'Excel. Français courant (C1) et anglais (C1).

COMPÉTENCES & MÉTHODES

Langages (prog.) : Python · R · SQL · SAS · Stata

Bureautique / BI : Excel (avancé) · PowerPoint · Power BI

Quantitatif : économétrie · modélisation & prévision · évaluation d'impact · analyse coûts-bénéfices

Spécialisation : régulation sectorielle · concurrence · obligations de service public · tarification & allocation des coûts · compensation / subventions

Langues : Français C1 · Espagnol (langue maternelle) · Anglais C1 · Catalan (notions)

EXPÉRIENCE

Spécialiste senior — Régulation & concurrence · Osinergmin (régulateur de l'énergie, Pérou) 2019–2024

- Défense d'intérêts dans plus de 50 litiges de concurrence : analyse économique quantifiée, pouvoir de marché et valorisation.
- Analyse quantitative et modélisation économique pour des procédures formelles devant les autorités de régulation.
- Communication pédagogique des résultats à la direction et aux comités techniques.

Spécialiste — Politiques & analyse économique · Osinergmin · Direction des politiques et de l'analyse économique 2014–2019

- Chiffrage économique et évaluation d'impact des obligations de service public et de leur compensation (dispositifs de subvention FOSE et MCTER).
- Conception tarifaire : structure des tarifs, allocation des coûts et besoins de revenus, en lien avec la comptabilité réglementaire.
- Modèles économiques de prévision et études d'impact ; analyse comparative internationale.

Spécialiste en politique macroéconomique · Ministère de l'Économie et des Finances (MEF), Pérou 2012–2013

- Modélisation et prévision macroéconomiques (modèles VAR) à l'appui de la décision publique.

FORMATION

Master en Économie, Régulation & Concurrence — Universitat de Barcelona (2025–2026)

Master en Data Science — La Salle – URL, Barcelone (2025–2026)

Master en Économie comportementale — Paris School of Economics (2018)

Master en Économie industrielle — Toulouse School of Economics (2005–2007)

Licence en Économie — PUCP, Pérou (2000–2004)

DOCUMENTS DE TRAVAIL

ÉLECTRICITÉ & MARCHÉS DE L'ÉNERGIE

1. The Price of Hedging: Forward Risk Premia in Iberian Power through the Energy Crisis, and What They Cost the Non-Integrated Retailer (2019–2024).

github.com/cmvddata/mibel-derivatives/blob/main/paper_derivatives/derivatives_paper.pdf

2. Screening the Firm Resource: De-rating Calibration, Adverse Selection and Lock-in in Spain's 2026/2027 Capacity Mechanism (SA.112483) — An Admissible-Space Audit with Hourly ELCC Data (2019–2024).
github.com/cmvddata/capacity/blob/main/paper/cm_spain_paper.pdf
3. Exit and Absorption in Spanish Electricity Retail through the Energy Crisis (2019–2024).
github.com/cmvddata/model_electricity/blob/main/retail_paper.pdf
4. No Firm Capacity, Three Markets: Solar and the Capacity Mechanism in Southern Europe.
github.com/cmvddata/southern-europe-renewables/blob/main/paper/three_markets.pdf
5. The Wrong Index: Concentration in the Binding Resource (Firm Capacity) in the Spanish Wholesale Market.
github.com/cmvddata/papers/blob/main/firm_concentration_paper.pdf

TRANSPORT

1. The Premium-Calibration Sensitivity of an Intermodal Pricing Ceiling: A Theory-Grounded Yield-Management Engine for the BCN–MAD Corridor.
github.com/cmvddata/papers/blob/main/yield_bcn_mad_paper.pdf
2. Where Did the Booking Fences Go? Advance-Purchase Pricing in a Post-Entry Air Duopoly on the Madrid–Barcelona Corridor.
github.com/cmvddata/papers/blob/main/pricing_paper.pdf
3. A Corridor Loses Both of Its Low-Cost Operators: Prices, Passengers, and a Registered Prediction.
github.com/cmvddata/papers/blob/main/avlo_preres.pdf
4. Pricing Response After the Low-Cost Brand Leaves: Independent High-Speed Fares on Madrid–Barcelona When AVLO Withdrew.
github.com/cmvddata/papers/blob/main/railprice_paper.pdf
5. Subsidising the Metro per Passenger or per Kilometre? A Calibrated Translog Cost Function for the Barcelona Metro (FMB-TMB), 2023–2025.
github.com/cmvddata/TMB-costos/blob/main/paper/paper_translog_fmb.pdf
6. The Generalised Price of the Madrid–Barcelona Corridor: What Realised Market Data Reveal About the Air–Rail Trade-off.
github.com/cmvddata/papers/blob/main/genprice_paper.pdf

DATA SCIENCE / MACHINE LEARNING

1. Detecting Structural Manipulation in Electricity Markets: A Supervised Approach Using Partial Regulatory Ground Truth.
github.com/cmvddata/neuro/blob/main/paper/main.pdf
2. Replicating LEAR on MIBEL through the Solar Transition: Why Technical Indicators Do Not Transfer, and Why Percentage Errors Mislead (2022–2024).
github.com/cmvddata/mibel-forecasting/blob/main/paper_forecasting/forecasting_paper.pdf
3. When Psychopathic Traits Are Not Traits: Aggregation Bias and Big-Five Redundancy in the Prediction of Professional Success.
github.com/cmvddata/psychopathy-success/blob/main/docs/paper.pdf
4. Cross-Sectional Characterization of High-Cost Healthcare Users in the US: How the Morbidity-Cost Gradient Varies by Insurance Type in MEPS 2022.
github.com/cmvddata/MEPS_analysis/blob/main/paper/meps_paper.pdf

ÉCONOMIE PUBLIQUE · SANTÉ · PSYCHOMÉTRIE

1. Energy Poverty and the Targeting of Electricity Subsidies: Evidence from the FOSE in Peru. (*mémoire de master*)
github.com/cmvddata/fose_analysis/blob/main/docs/paper_fose.pdf
2. Targeting Energy Poverty: Structural Predictors versus Income Thresholds in Spain's Social Electricity Bonus.
github.com/cmvddata/bono-social-microsim/blob/main/paper/paper.pdf
3. Ordinal Distortion of the Spanish Regional Financing System: Quantitative Evidence for the Common-Regime Autonomous Communities, 2009–2022.
github.com/cmvddata/sfa-distorsion-ordinal/blob/main/paper/paper.pdf

CULTURE & SPORT

1. The Red and the Black: Colombia, Stendhal, and the Price of Arriving Too Soon.
github.com/cmvddata/culture-sports/blob/main/red_and_the_black.pdf
2. Who Drew the Line? Barcelona '92, the Dream Team, and the Industrial Organization of Painted Lines.

github.com/cmvddata/culture-sports/blob/main/who_drew_the_line.pdf

3. You Can't ABX a Memory: New Order's "Bizarre Love Triangle," Glastonbury 1987 vs 2016.
github.com/cmvddata/culture-sports/blob/main/you_cant_abx_a_memory.pdf
4. Was Cumbia Villera Invented in a Yamaha Office in Hamamatsu, Japan? An Aesthetics of Limitation: Technological Democratization, the "Fertile Error," and Sonic Identity in Cumbia Villera.
github.com/cmvddata/culture-sports/blob/main/cumbia_villera_hamamatsu.pdf
5. What Does a Country Buy When It Buys Back a Ball? Relics, Liturgy, and Sixty Years of Hurt: the 1966 Ball and the Symbolic Economy of English Football.
github.com/cmvddata/culture-sports/blob/main/buying_back_a_ball.pdf
6. Mexico '86, Marx and the Hand of God: How a Symbol Makes a Price and an Enemy in the Same Gesture.
github.com/cmvddata/culture-sports/blob/main/mexico86_hand_of_god.pdf
7. At the End, Totò Always Scores: Chance, Merit, and the Italian Summer That Chose Schillaci.
github.com/cmvddata/culture-sports/blob/main/at_the_end_toto_always_scores.pdf
8. Answered to Its Face: The Haka and the Open Game.
github.com/cmvddata/culture-sports/blob/main/haka_open_game.pdf
9. W/kg on a Climb is a Distribution, not a Verdict: Estimating Cycling Climbing Power with Honest Uncertainty.
github.com/cmvddata/cycling/blob/main/wkg/paper.pdf
10. Relaxing the Offside Law in Association Football: A Game-Theoretic, Agent-Based Simulation of Attack–Defense Equilibria.
github.com/cmvddata/offside/blob/main/offside_paper.pdf

PUBLICATIONS & INSTITUTIONNEL

Co-éditeur d'un ouvrage d'Osinergmin sur le secteur électrique péruvien (2016) et co-auteur de deux documents de travail sur la régulation du secteur de l'énergie (Working Papers 32 et 40, 2014–2016). Références disponibles sur demande.